

You have to design a POS tagger for Hindi. You are given the POS tagged files with train, dev, and test splits. Implement the following algorithms.

1. Naive Bayes Classifier: You need to predict the tag of each word. The features for each word are:
 - a. Length of a word: It is a binary feature. If length of the word > 4 , then the length feature is "MORE" else "LESS"
 - b. Actual Word
 - c. Previous Word
 - d. Next Word
 - e. Previous Tag
2. Design an HMM to predict the tag sequence given a sentence
 - a. Implement the standard forward Viterbi
 - b. Implement the backward Viterbi
 - c. Implement the forward-backward Viterbi

Calculate the probabilities from the train data and evaluate them on dev and test data. Report the F1-score for each tag. Compare the performance of all algorithms. Plot charts to compare the performance on each tag.